Survey Workbench

User Manual

October 2026

Version 2.2

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1 Overview

The Survey Workbench is a robust desktop application designed to streamline participant folder generation and automate questionnaire data extraction from fillable PDF forms into centralized CSV or Excel masterfiles. Version 2.2 features a streamlined interface with a collapsible configuration container, dynamic window resizing, and robust configuration management.

1.1 Key Features

- Collapsible Configuration container (hidden by default for cleaner end-user presentation)
- Dynamic window resizing that adapts automatically to presented content
- Dynamic questionnaire configuration supporting unlimited custom survey types
- Batch processing capabilities for generating folders and extracting data across multiple participants simultaneously
- Participant list import functionality supporting text and CSV formats
- Configuration snapshotting for saving and reloading complete study setups, folder paths, and parameters
- Automated duplicate detection against masterfiles (CSV, XLS, XLSX)
- Data completeness verification and missing data reporting
- Interactive help system featuring tooltips and a dedicated "What's This?" mode
- Automated format detection for masterfiles with interactive preview dialogs prior to committing data

1.2 System Requirements

- Windows 10/11 or Linux operating systems
- Microsoft Excel (required for Excel masterfile readout via xlwings on Windows)
- Python 3.x environment with PyQt5 and pypdf dependencies (or deployed as a standalone compiled binary)

2 Getting Started and Interactive Help

When you first launch Survey Workbench v2.2, you will see a clean, focused window containing:

1. **Participant Declaration** – For entering single or multiple participant IDs.
2. **Configuration Settings** – A collapsible dropdown container (collapsed by default) for advanced setup.
3. **Action Buttons** – Always visible buttons to generate participant folders or extract data immediately.

If you are ever unsure what a specific control does:

- **Tooltips**: Hover your mouse over any control to see a brief description of its function.
- **What's This? Mode**: Navigate to **HELP → What's This? (Click for Help)** or press **Shift+F1**. Your cursor will change into a question mark; click on any interface element to open a detailed explanatory window outlining its exact purpose and usage.

3 Configuration Settings

For study curators, click the **Configuration Settings** toggle button to reveal the configuration panels. The window will dynamically expand to fit the settings.

3.1 Target and Source Paths

- **Target folder PATH**: Choose where participant folders will be created during generation.
- **Source folder PATH**: Choose the folder containing completed participant data folders.
- **Masterfile PATH**: Choose your masterfile (.csv, .xls, or .xlsx) for data readout.

3.2 Configuring Questionnaire Types

1. Enter the number of different questionnaire types in **Number of questionnaire types**.
2. Click **Confirm** to generate the scrollable configuration rows.
3. For each row, specify the **Survey Name**, select the PDF template via **Search**, and set the **Number of copies**.

4 Step-by-Step Guide: Generating Participant Folders

4.1 1. Declaring Participant IDs

- **Single Mode**: Enter a unique identifier directly into the **Participant ID** text box (e.g., `P001`). This will be used as the subfolder name.
- **Batch Mode**: Check **Batch Mode (multiple IDs)** to open a text area where you can input multiple participant identifiers (one per line or comma-separated). Alternatively, click **Import List...** to load IDs directly from an external text or CSV file.

4.2 2. Generating Folders

Click **Generate Participant Folder(s)**. Even when the configuration settings are collapsed, the system uses your active configuration in the background to create subfolders for each participant ID inside the target folder, copy and rename questionnaire templates according to your specifications, and open the consolidated prefill dialog to review and edit form data before final output.

5 Step-by-Step Guide: Extracting Data to Masterfiles

With the participant ID(s) declared and source/masterfile paths configured, click **Extract Data to Masterfile**.

5.1 Batch and Single Extraction

- For single participants, a 4-column preview dialog lets you inspect and edit field mapping names and extracted data values prior to commit.
- For multiple participants or batch mode, data is processed across all specified participants in a single consolidated session, writing directly into dedicated sheets per root PDF in Excel or appending rows to CSV masterfiles.
- Column A in Excel sheets is automatically designated as `participant_id`.

5.2 Quality Control: Completeness and Duplicates

Before running extraction, the software incorporates built-in safety checks:

- **Duplicate Detection**: Automatically scans the masterfile to check if a participant ID has already been extracted, preventing accidental duplicate entries.
- **Data Completeness**: Verifies that all expected readable PDF form fields and data files are present. You can also click **Missing Data Report** to scan the entire source directory and generate a comprehensive summary of complete versus incomplete participant folders.

5.3 Field Mapping

Access **File** → **Save/Edit Field Mapping** from the menu bar to map technical PDF form field identifiers to user-friendly column headers. Mappings are automatically saved into your active configuration snapshot.

6 Configuration Management

Save and load complete snapshots of your study setupincluding folder paths, questionnaire parameters, and custom field mappingsusing **File** → **Save Configuration** and **File** → **Load Configuration**. Saved configurations can be easily reloaded or deleted using the respective submenus under the **File** menu.

7 Technical Information and Version History

7.1 Complete Version History

- **v2.2 (Current Edition)**: Introduced collapsible curator configuration containers, dynamic window resizing, comprehensive PDF form field extraction via AcroForm and widget annotation scanning, interactive prefill and extraction preview dialogs, robust batch processing with list imports, automated duplicate checking, missing data reporting, and flexible configuration snapshotting.
- **v2.1**: Added structured field mapping capabilities, enhanced masterfile format autodetection, and refined prefill preset handling.
- **v2.0**: Established dynamic questionnaire configuration supporting unlimited questionnaire types and unified masterfile readout for CSV and Excel formats.
- **v1.0 (Initial Release)**: Basic static folder generation and single-format CSV data extraction for fixed questionnaire types (Demographics and NASA-TLX).

7.2 Technology Stack and Deployment

- Python 3.x
- PyQt5 (GUI framework featuring tooltips and What's This interactive help)
- xlwings (Excel integration and workbook automation)
- pypdf (PDF form field reading and writing)
- Nuitka (Cross-platform compiler used to compile the source code into native standalone executables, generating '.exe' binaries on Windows and ELF binaries on Linux)